

$$y = (mc - nb)f(x) - (lc - na)g(x) + (lb - ma)h(x)$$

$$\Rightarrow \frac{dy}{dx} = (mc - nb)f'(x) - (lc - na)g'(x) + (lb - ma)h'(x)$$

$$= \begin{vmatrix} f'(x) & g'(x) & h'(x) \\ l & m & n \\ a & b & c \end{vmatrix}$$

Multiple Choice Questions

Choose and write the correct option in the following questions.

- The function $f: \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = -|x - 1|$ is [CBSE 2020 (65/2/1)]
 - continuous as well as differentiable at $x = 1$
 - not continuous but differentiable at $x = 1$
 - continuous but not differentiable at $x = 1$
 - neither continuous nor differentiable at $x = 1$
- The function $f(x) = e^{|x|}$ is [NCERT Exemplar]
 - continuous everywhere but not differentiable at $x = 0$
 - continuous and differentiable everywhere
 - not continuous at $x = 0$
 - none of these
- The function $f(x) = [x]$, where $[x]$ denotes the greatest integer function, is continuous at
 - 4
 - 2
 - 1
 - 1.5
- The number of points at which the function $f(x) = \frac{1}{x - [x]}$ is not continuous is [NCERT Exemplar] [Competency Based Question]
 - 1
 - 2
 - 3
 - none of these
- The function $f(x) = \begin{cases} \frac{\sin x}{x} + \cos x, & \text{if } x \neq 0 \\ k, & \text{if } x = 0 \end{cases}$ is continuous at $x = 0$, then the value of k is [NCERT Exemplar] [Competency Based Question]
 - 3
 - 2
 - 1
 - 1.5
- If the function f defined by $f(x) = \begin{cases} \frac{k \cos x}{\pi - 2x} & \text{if } x \neq \frac{\pi}{2} \\ 3, & \text{if } x = \frac{\pi}{2} \end{cases}$ is continuous at $x = \frac{\pi}{2}$, then the value of k is [CBSE Term-1, 2021-22 (65/2/4)]
 - 2
 - 3
 - 6
 - 6
- The function $f(x) = \cot x$ is discontinuous on the set [NCERT Exemplar]
 - $\{x = n\pi : n \in \mathbb{Z}\}$
 - $\{x = 2n\pi : n \in \mathbb{Z}\}$
 - $\left\{x = (2n + 1)\frac{\pi}{2} : n \in \mathbb{Z}\right\}$
 - $\left\{x = \frac{n\pi}{2} : n \in \mathbb{Z}\right\}$
- Let $f(x) = |\sin x|$. Then [NCERT Exemplar] [Competency Based Question]
 - f is everywhere differentiable.
 - f is everywhere continuous but not differentiable at $x = n\pi : n \in \mathbb{Z}$.
 - f is everywhere continuous but not differentiable at $x = (2n + 1)\frac{\pi}{2}, n \in \mathbb{Z}$.
 - none of these

9. The function $f(x) = \frac{x-1}{x(x^2-1)}$ is discontinuous at [CBSE 2020 (65/2/2)]
 (a) exactly one point (b) exactly two points
 (c) exactly three points (d) no point
10. If $f(x) = x^2 \sin \frac{1}{x}$, where $x \neq 0$, then the value of the function f at $x = 0$, so that the function is continuous at $x = 0$, is [NCERT Exemplar] [Competency Based Question]
 (a) 0 (b) -1 (c) 1 (d) None of these
11. The function $f(x) = \begin{cases} \frac{e^{3x} - e^{-5x}}{x}, & \text{if } x \neq 0 \\ 3, & \text{if } x = 0 \end{cases}$, is continuous at $x = 0$ for the value of k , as [CBSE Term-1, 2021-22 (65/4/2)]
 (a) 3 (b) 5 (c) 2 (d) 8
12. If $y = \sin(m \sin^{-1}x)$, then which of the following equations is true? [CBSE Term-1, 2021-22 (65/2/4)]
 (a) $(1-x^2)\frac{d^2y}{dx^2} + x\frac{dy}{dx} + m^2y = 0$ (b) $(1-x^2)\frac{d^2y}{dx^2} - x\frac{dy}{dx} + m^2y = 0$
 (c) $(1+x^2)\frac{d^2y}{dx^2} - x\frac{dy}{dx} - m^2y = 0$ (d) $(1+x^2)\frac{d^2y}{dx^2} + x\frac{dy}{dx} - m^2y = 0$
13. Differential of $\log [\log (\log (x^5))]$ wrt x , is [CBSE Term-1, 2021-22 (65/2/4)]
 (a) $\frac{5}{x \log(x^5) \log(\log x^5)}$ (b) $\frac{5}{x \log(\log x^5)}$
 (c) $\frac{5x^4}{\log(x^5) \log(\log x^5)}$ (d) $\frac{5x}{\log(x^5) \log(\log x^5)}$
14. If $\sin y = x \cos(a+y)$, then $\frac{dx}{dy}$ is [CBSE Term-1, 2021-22 (65/2/4)]
 (a) $\frac{\cos a}{\cos^2(a+y)}$ (b) $\frac{-\cos a}{\cos^2(a+y)}$ (c) $\frac{\cos a}{\sin^2 y}$ (d) $\frac{-\cos a}{\sin^2 y}$
15. If $(x^2 + y^2)^2 = xy$, then $\frac{dy}{dx}$ is [CBSE Term-1, 2021-22 (65/2/4)]
 (a) $\frac{y + 4x(x^2 + y^2)}{4y(x^2 + y^2) - x}$ (b) $\frac{y - 4x(x^2 + y^2)}{x + 4(x^2 + y^2)}$ (c) $\frac{y - 4x(x^2 + y^2)}{4y(x^2 + y^2) - x}$ (d) $\frac{4y(x^2 + y^2) - x}{y - 4x(x^2 + y^2)}$
16. The set of points where the functions f given by $f(x) = |x-3| \cos x$ is differentiable is
 (a) $\mathbb{R}$ (b) $\mathbb{R} - \{3\}$ (c) $(0, \infty)$ (d) none of these
17. Differential coefficient of $\sec (\tan^{-1}x)$ w.r.t. x is [NCERT Exemplar]
 (a) $\frac{x}{\sqrt{1+x^2}}$ (b) $\frac{x}{1+x^2}$ (c) $x\sqrt{1+x^2}$ (d) $\frac{1}{\sqrt{1+x^2}}$
18. If $u = \sin^{-1}\left(\frac{2x}{1+x^2}\right)$ and $v = \tan^{-1}\left(\frac{2x}{1-x^2}\right)$, then $\frac{du}{dv}$ is [Competency Based Question]
 (a) $\frac{1}{2}$ (b) x (c) $\frac{1-x^2}{1+x^2} \{4, -4\}, \phi$ (d) 1
19. If $y = \log \sqrt{\tan x}$, then the value of $\frac{dy}{dx}$ at $x = \frac{\pi}{4}$ is [Competency Based Question]
 (a) 0 (b) 1 (c) $\frac{1}{2}$ (d) ∞
20. If $y = \sqrt{\sin x + y}$, then $\frac{dy}{dx}$ is equal to [Competency Based Question]
 (a) $\frac{\cos x}{2y-1}$ (b) $\frac{\cos x}{1-2y}$ (c) $\frac{\sin x}{1-2y}$ (d) $\frac{\sin x}{2y-1}$

Answers

- | | | | | | | |
|---------|---------|---------|---------|---------|---------|---------|
| 1. (c) | 2. (a) | 3. (d) | 4. (d) | 5. (b) | 6. (c) | 7. (a) |
| 8. (b) | 9. (c) | 10. (a) | 11. (d) | 12. (b) | 13. (a) | 14. (a) |
| 15. (c) | 16. (b) | 17. (a) | 18. (d) | 19. (b) | 20. (a) | |

Solutions of Selected Multiple Choice Questions

1. We have,

$$f(x) = -|x-1| = \begin{cases} x-1, & \text{if } x \leq 1 \\ -(x-1), & \text{if } x > 1 \end{cases}$$

At $x = 1$

$$\text{LHL} = \lim_{h \rightarrow 0} f(1-h) = \lim_{h \rightarrow 0} (1-h) - 1 = 0$$

$$\text{RHL} = \lim_{h \rightarrow 0} f(1+h) = \lim_{h \rightarrow 0} -(1+h-1) = 0$$

$$f(1) = 1 - 1 = 0$$

$\therefore \text{LHL} = \text{RHL} = f(1) \Rightarrow f(x)$ is continuous every where.

Now, at $x = 1$

$$\text{LHD} = \frac{d}{dx}(x-1) = 1; \quad \text{RHD} = \frac{d}{dx}\{-(x-1)\} = -1$$

$$\text{LHD} \neq \text{RHD}$$

$\therefore f(x)$ is not differentiable of $x = 1$.

6. Given $f(x)$ is continuous at $x = \frac{\pi}{2}$,

$$\therefore \lim_{x \rightarrow \frac{\pi}{2}} f(x) = f\left(\frac{\pi}{2}\right)$$

$$\begin{aligned} \lim_{x \rightarrow \frac{\pi}{2}} f(x) &= \lim_{x \rightarrow \frac{\pi}{2}} \left(\frac{k \cos x}{\pi - 2x} \right) \\ &= \lim_{x \rightarrow \frac{\pi}{2}} \frac{-k \sin x}{-2} = \lim_{x \rightarrow \frac{\pi}{2}} \frac{k \sin x}{2} \\ &= \frac{k}{2} \sin \frac{\pi}{2} = \frac{k}{2} \end{aligned}$$

$$\therefore f\left(\frac{\pi}{2}\right) = 3$$

$$\therefore \lim_{x \rightarrow 0} f(x) = f\left(\frac{\pi}{2}\right)$$

$$\Rightarrow \frac{k}{2} = 3 \Rightarrow k = 6$$

7. We know that, $f(x) = \cot x$ is continuous in $R - \{n\pi : n \in Z\}$.

$$\text{Since, } f(x) = \cot x = \frac{\cos x}{\sin x} \quad [\text{since, } \sin x = 0 \text{ at } \{n\pi, n \in Z\}]$$

Hence, $f(x) = \cot x$ is discontinuous on the set $\{x = n\pi : n \in Z\}$.

9. We have,

$$f(x) = \frac{x-1}{x(x^2-1)}$$

$\therefore f(x)$ is discontinuous when $x(x^2-1) = 0$

$$\Rightarrow x = 0, x = \pm 1$$

$\therefore f(x)$ is discontinuous at $x = 0, -1, 1$
i.e., exactly at three points.

10. $\because f(x) = x^2 \sin\left(\frac{1}{x}\right)$, where $x \neq 0$; $\therefore \lim_{x \rightarrow 0} f(x) = 0$

Hence, value of the function f at $x = 0$, so that it is continuous at $x = 0$ is 0.

11. Given function is continuous at $x = 0$.

$$\therefore \lim_{x \rightarrow 0} f(x) = f(0)$$

$$\begin{aligned} \therefore \lim_{x \rightarrow 0} f(x) &= \lim_{x \rightarrow 0} \left(\frac{e^{3x} - e^{-5x}}{x} \right) \\ &= \lim_{x \rightarrow 0} \left(\frac{3e^{3x} + 5e^{-5x}}{1} \right) = 3 + 5 = 8 \end{aligned}$$

$$\therefore f(0) = k$$

$$\Rightarrow k = 8$$

12. Given $y = \sin(m \sin^{-1} x)$

$$\therefore \frac{dy}{dx} = \cos(m \sin^{-1} x) \times \frac{m}{\sqrt{1-x^2}} = \frac{m \cos(m \sin^{-1} x)}{\sqrt{1-x^2}}$$

$$\begin{aligned} \therefore \frac{d^2y}{dx^2} &= m \left\{ \frac{\sqrt{1-x^2} \times -\sin(m \sin^{-1} x) \times \frac{m}{\sqrt{1-x^2}} - \cos(m \sin^{-1} x) \times \frac{1}{2\sqrt{1-x^2}} \times (-2x)}{(\sqrt{1-x^2})^2} \right\} \\ &= \frac{-m^2 \sin(m \sin^{-1} x) + \frac{xm \cos(m \sin^{-1} x)}{\sqrt{1-x^2}}}{1-x^2} \end{aligned}$$

$$\Rightarrow (1-x^2) \frac{d^2y}{dx^2} = -m^2 \sin(m \sin^{-1} x) + x \frac{dy}{dx}$$

$$\Rightarrow (1-x^2) \frac{d^2y}{dx^2} = -m^2 y + x \frac{dy}{dx} = -m^2 y + x \frac{dy}{dx}$$

$$\Rightarrow (1-x^2) \frac{d^2y}{dx^2} - x \frac{dy}{dx} + m^2 y = 0$$

13. Let $y = \log [\log (\log (x^5))]$

By using chain rule, we have

$$\begin{aligned} \frac{dy}{dx} &= \frac{d}{dx} \log [\log (\log x^5)] \\ &= \frac{1}{\log (\log x^5)} \times \frac{1}{\log (x^5)} \times \frac{1}{x^5} \times 5x^4 = \frac{5}{x \log (x^5) \log (\log x^5)} \end{aligned}$$

14. $\because \sin y = x \cos(a+y)$

$$\Rightarrow \frac{\sin y}{\cos(a+y)} = x \quad \Rightarrow \quad x = \frac{\sin y}{\cos(a+y)}$$

$$\begin{aligned} \therefore \frac{dx}{dy} &= \frac{\cos(a+y) \cos y - \sin y \times -\sin(a+y)}{\cos^2(a+y)} \\ &= \frac{\cos(a+y) \cos y + \sin(a+y) \sin y}{\cos^2(a+y)} \\ &= \frac{\cos(a+y-y)}{\cos^2(a+y)} = \frac{\cos a}{\cos^2(a+y)} \end{aligned}$$

15. $\because (x^2 + y^2)^2 = xy$

Differentiate wrt x , we get

$$\begin{aligned} 2(x^2 + y^2) \left(2x + 2y \frac{dy}{dx} \right) &= x \frac{dy}{dx} + y \\ \Rightarrow 4x(x^2 + y^2) + 4y(x^2 + y^2) \frac{dy}{dx} &= \frac{xdy}{dx} + y \\ \Rightarrow \{ 4y(x^2 + y^2) - x \} \frac{dy}{dx} &= y - 4x(x^2 + y^2) \\ \Rightarrow \frac{dy}{dx} &= \frac{y - 4x(x^2 + y^2)}{4y(x^2 + y^2) - x} \end{aligned}$$

19. We have, $y = \log \sqrt{\tan x}$

$$\begin{aligned} \therefore \frac{dy}{dx} &= \frac{1}{\sqrt{\tan x}} \times \frac{1}{2\sqrt{\tan x}} \times \sec^2 x \\ \Rightarrow \frac{dy}{dx} &= \frac{\sec^2 x}{2 \tan x} \Rightarrow \frac{dy}{dx} \text{ at } x = \frac{\pi}{4} = \frac{(\sqrt{2})^2}{2 \times 1} = \frac{2}{2} = 1 \end{aligned}$$

20. $\because y = (\sin x + y)^{1/2}$

$$\begin{aligned} \frac{dy}{dx} &= \frac{1}{2} (\sin x + y)^{-1/2} \cdot \frac{d}{dx} (\sin x + y) = \frac{1}{2} \cdot \frac{1}{(\sin x + y)^{1/2}} \cdot \left(\cos x + \frac{dy}{dx} \right) \\ \Rightarrow \frac{dy}{dx} &= \frac{1}{2y} \left(\cos x + \frac{dy}{dx} \right) \quad [\because (\sin x + y)^{1/2} = y] \\ \Rightarrow \frac{dy}{dx} \left(1 - \frac{1}{2y} \right) &= \frac{\cos x}{2y} \Rightarrow \frac{dy}{dx} = \frac{\cos x}{2y} \cdot \frac{2y}{2y - 1} = \frac{\cos x}{2y - 1} \end{aligned}$$

Very Short Answer Questions

Each of the following questions are of 1 mark.

1. If the function f defined as $f(x) = \begin{cases} x^2 - 9, & x \neq 3 \\ k, & x = 3 \end{cases}$ is continuous at $x = 3$, find the value of k . [CBSE 2020 (65/5/1)]

Sol. $\because$ It is given that $f(x)$ is continuous at $x = 3$.

$$\begin{aligned} \therefore \lim_{x \rightarrow 3} f(x) &= f(3) \\ \Rightarrow \lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3} &= k \Rightarrow \lim_{x \rightarrow 3} \frac{(x - 3)(x + 3)}{(x - 3)} = k \\ \Rightarrow \lim_{x \rightarrow 3} (x + 3) &= k \Rightarrow 3 + 3 = k \Rightarrow k = 6 \end{aligned}$$

2. For what value of ' k ' is the function $f(x) = \begin{cases} \frac{\sin 5x}{3x} + \cos x, & \text{if } x \neq 0 \\ k, & \text{if } x = 0 \end{cases}$ continuous at $x = 0$? [CBSE (F) 2017]

Sol. $\lim_{x \rightarrow 0^+} f(x) = \lim_{h \rightarrow 0} f(0 + h)$

$$\begin{aligned} &= \lim_{h \rightarrow 0} f(h) = \lim_{h \rightarrow 0} \left(\frac{\sin 5h}{3h} + \cos h \right) \\ &= \lim_{h \rightarrow 0} \frac{\sin 5h}{5h} \times \frac{5}{3} + \lim_{h \rightarrow 0} \cos h = 1 \times \frac{5}{3} + 1 \quad [\because h \rightarrow 0 \Rightarrow 5h \rightarrow 0] \end{aligned}$$